1 Which group of quantities contains only vectors?

A acceleration, displacement, speed

B acceleration, work, electric field strength

C displacement, force, velocity

D power, electric field strength, force

2 Which statement using prefixes of the base unit metre (m) is **not** correct?

A $1\,\text{pm} = 10^{-12}\,\text{m}$

B $1\,\text{nm} = 10^{-9}\,\text{m}$

C $1\,\text{Mm} = 10^{6}\,\text{m}$

D $1\,\text{Gm} = 10^{12}\,\text{m}$

3 A current of (2.50 ± 0.05) mA passes through a resistor of resistance $(4.7 \pm 0.2)\Omega$.

What is the percentage uncertainity in the power dissipated in the resistor?

A 2%
B 4%
C 6%
D 8%

4

Which of the following quantities has base units $kgm^{-1}s^{-2}$?

A pressure
B energy
C force
D momentum

5

Which phrase describes an accurate reading?

A low random error
B high precision
C low precision
D low systematic error

6

D volt per metre

If p is the momentum of a body of mass m, which physical quantity has the same base units as the expression $\frac{p^2}{m}$?

A energy
B force
C impulse
D power

7.

Which equation is **not** homogenous where symbols have their usual meanings?

A $k.e = \frac{1}{2}mv^2$

B $T = 2\pi\sqrt{g/l}$

C $a = \omega^2 r$

D $s = ut + at^2$

8.

Which set of experimental results for the acceleration of free fall, g, could be precise but **not** accurate?

A 9.81; 9.79; 9.84; 9.83 9,8145
B 9.81; 10.12; 9.89; 8.94
C 9.45; 9.21; 8.99; 8.76
D 8.45; 8.46; 8.50; 8.41

9.

Which experimental results are precise but **not** accurate for the determination of the acceleration of free fall in ms^{-2}?

A 8.10; 8.02; 8.07
B 8.90; 8.99; 9.81
C 9.45; 9.81; 9.79
D 9.80; 9.79; 9.81

10

What is the greatest height reached by a ball thrown vertically into free space with an initial velocity u?

A $\frac{u^2}{g}$

B $\frac{u}{2g}$

C $\frac{3u^2}{2g}$

D $\frac{u^2}{2g}$

11.

Which pair includes a vector quantity and a scalar quantity?

A displacement; acceleration

B force; kinetic energy

C power; speed

D work; potential energy

12

What does the prefix *tera* represent?

A 10^{15}

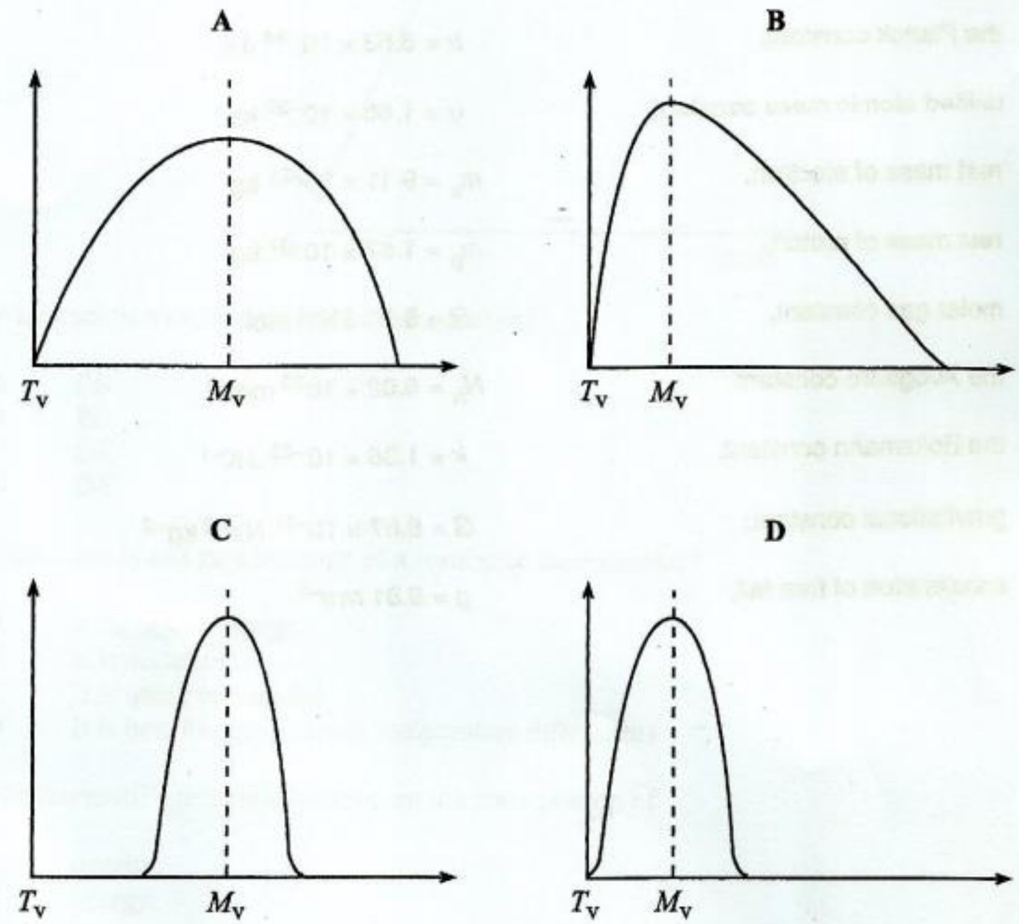
B 10^{12}

C 10^{9}

D 10^{6}

13

Which graph shows a large systematic error and a small random error given that T_V is the true and M_V is the mean value?



14

Systematic errors can

A be reduced by repeating measurements.
B be reduced by changing the instrument.
C only be due to wrong assumptions.
D cause readings to scatter around the true value.

15

Which statement does **not** show that there is a resultant force acting on a particle?

A The particle is moving.
B The particle is turning.
C The particle is accelerating.
D The particle is free falling.

16

A stunt rider leaves a ramp at an angle of 30° above the horizontal and lands at the same height, *h*, after travelling a horizontal distance of 36 m.

What was the take off speed?

A $10\ ms^{-1}$
B $12\ ms^{-1}$
C $14\ ms^{-1}$
D $20\ ms^{-1}$

30

$R = 36$

$x = u\cos\theta t$

$36 = u\cos 30 t$

$h = u\sin\theta t - \frac{1}{2}gt^2$

$\frac{24\sqrt{3}}{u} = t$

$u\sin\theta\left(\frac{24\sqrt{3}}{u}\right) - \ldots\left(\frac{24\sqrt{3}}{u}\right)$

17.

Which of the following has the same units as momentum?

A force
B pressure
C the product of force and time
D product of force and velocity

mv

$kg\,ms^{-1}$

$kg\,ms^{-1}$

18

Which phrase describes an accurate reading?

A low random error
B high precision
C low precision
D low systematic error

19

Which statement is correct about the resultant force acting on a mass undergoing uniform acceleration?

A It increases uniformly with respect to time.
B It is constant but not zero.
C It is proportional to the displacement from a fixed point.
D It is proportional to the velocity.

20.

Which quantity remains constant for an orange falling from a tree?

A velocity
B displacement
C acceleration
D speed

21

Which definition is **not** correct?

A Acceleration is the rate of change of displacement with time.
B Force is the rate of change of momentum.
C Impulse is the change in momentum.
D Linear momentum is the product of mass and velocity.

22

A car uniformly decelerates from 30 ms^{-1} to 15 ms^{-1} over a distance of 75 m.

What total distance is moved by the car before coming to rest?

A 25 m
B 75 m
C 100 m
D 175 m

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